Federico **Nocentini**

Postdoctoral Research Fellow
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Google Scholar GitHub LinkedIn Personal page

Summary

Machine Learning researcher with a Ph.D. in 3D Computer Vision and Artificial Intelligence, specializing in generative models for 3D human animation and speech-driven facial animation. Experienced in conducting cutting-edge research independently and as part of collaborative teams in both academia and industry, with a proven track record of publishing in top-tier venues such as ECCV, WACV, IJCV and FG. Proficient in Python and PyTorch, with hands-on expertise in diffusion models and 3D geometry processing, as demonstrated by open-source contributions. Strong analytical, problem-solving and communication skills. Highly motivated to contribute to innovative projects and to apply expertise in 3D vision and generative AI to both exploratory and applied research settings.

Experience

- **Postdoctoral Research Fellow** — **Dublin, Ireland**
  Trinity College Dublin, Graphics and Vision Group — *June 2026 - Present*

- **Research Scientist Intern** — **Los Angeles, California**
  Flawless AI — *November 2024 - May 2025*
  Multilingual speech-driven facial animation.

Education

- **PhD in 3D Computer Vision and Artificial Intelligence** — **Florence, Italy**
  University of Florence - Media Integration and Communication Center — *November 2022 - April 2026*
  Thesis: 3D Talking Heads: Advancing Realism and Generalization.
  Advisors: Prof. Stefano Berretti, Prof. Claudio Ferrari.

- **M.Sc in Computer Science and Engineering** — **Florence, Italy**
  University of Florence, Grade: 110/110 (GPA: 4.0/4.0) — *February 2020 - October 2022*
  Thesis: Generating Emotional 3D Talking Heads from Audio Spectrograms.

- **B.Sc in Computer Science and Engineering** — **Florence, Italy**
  University of Florence, Grade: 104/110 (GPA: 3.8/4.0) — *September 2016 - February 2020*
  Thesis: Facial recognition with depth images generated from a deformable model of the face.

Selected Publications

A comprehensive publication list is available on my Google Scholar profile.

- **F. Nocentini**, T. Besnier, C. Ferrari, S. Berretti, M. Daoudi, **FreeTalk: Emotional Topology-Free 3D Talking Heads**, **arXiv**, 2026
- **F. Nocentini**, K. Seo, Q. Liu, C. Ferrari, S. Berretti, D. Ferman, H. Kim, P. Garrido, A. Caliskan, **Polyglot: Multilingual Style Preserving Speech-Driven Facial Animation**, IEEE International Conference on Automatic Face and Gesture Recognition (**FG**), 2026
- **F. Nocentini**, T. Besnier, C. Ferrari, S. Arguillere, M. Daoudi, S. Berretti, **Beyond Fixed Topologies: Unregistered**

Training and Comprehensive Evaluation Metrics for 3D Talking Heads, International Journal of Computer Vision (**IJCV**), 2026

- E. Sariyanidi, C. Ferrari, **F. Nocentini**, S. Berretti, A. Cavallaro, B. Tunc, **3D Face Reconstruction Error Decomposed: A Modular Benchmark for Fair and Fast Method Evaluation**, IEEE International Conference on Automatic Face and Gesture Recognition (**FG**), 2025
- **F. Nocentini**, C. Ferrari, S. Berretti, **EmoVOCA: Speech-Driven Emotional 3D Talking Heads**, IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**), 2025
- **F. Nocentini**, T. Besnier, C. Ferrari, S. Arguillere, S. Berretti, M. Daoudi, **ScanTalk: 3D Talking Heads from Unregistered Scans**, European Conference on Computer Vision (**ECCV**), 2024
- **F. Nocentini**, C. Ferrari, S. Berretti, **Learning Landmarks Motion from Speech for Speaker-Agnostic 3D Talking Heads Generation**, International Conference on Image Analysis and Processing (**ICIAP**), 2023

Awards

- **Best Demo Award**, IEEE International Joint Conference on Biometrics (**IJCB**), 2023, for *S2L+S2D: Identity-Preserving Speech-Driven 3D Talking Heads*

Languages

- **Italian**: Mother tongue
- **English**: Full professional proficiency

Skills

- **Frameworks**: PyTorch, Tensorflow, Open3D, PyTorch3D, Trimesh, OpenCV, Numpy, Pandas, Matplotlib
- **Programming**: Python, C/C++, CUDA, OpenMP, Java, JavaScript